

Experiment - 10

Student Name: Kamal Sharma
Branch: CSE - AIML
Semester: 5
Subject Name: ADBMS

UID: 24BAI70380
Section/Group: 24AIT_KRG-1_B
Date of Performance: 14-09-26
Subject Code: 24CSP-310

(A)

Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction. If any insert fails due to invalid data, only that insert should be rolled back while preserving the previous successful inserts using savepoints. The system should provide clear messages for both successful and failed insertions, ensuring data integrity and controlled error handling.

Solution :

```
CREATE OR REPLACE PROCEDURE insert_students()
LANGUAGE PLPGSQL
AS $$
BEGIN
    -- Insert 1
    BEGIN
        INSERT INTO students(name, age, class)
        VALUES ('Anisha', 16, 8);
        RAISE NOTICE 'Anisha inserted successfully';
    EXCEPTION WHEN OTHERS THEN
        RAISE NOTICE 'Anisha insertion failed. Rolled back.';
    END;
```

-- Insert 2

BEGIN

INSERT INTO students(name, age, class)

VALUES ('Neha', 17, 8);

RAISE NOTICE 'Neha inserted successfully';

EXCEPTION WHEN OTHERS THEN

RAISE NOTICE 'Neha insertion failed. Rolled back.';

END;

-- Insert 3 INVALID

BEGIN

INSERT INTO students(name, age, class)

VALUES ('Anisha', 19, 9);

RAISE NOTICE 'Anisha inserted successfully';

EXCEPTION WHEN OTHERS THEN

RAISE NOTICE 'Anisha insertion failed. Only this insertion was rolled
back.';

END;

-- Insert 4

BEGIN

INSERT INTO students(name, age, class)

VALUES ('Mayank', 19, 9);

RAISE NOTICE 'Mayank inserted successfully';

EXCEPTION WHEN OTHERS THEN

RAISE NOTICE 'Mayank insertion failed. Rolled back.';

END;

```
RAISE NOTICE 'Transaction Successfully Done';  
END;  
$$;
```

```
BEGIN;  
CALL insert_students();  
COMMIT;
```

```
SELECT * FROM students;
```